Project Logbook

Evaluating the Potential of Zinc-ion Batteries

Goal

Tuesday, August 20, 2024

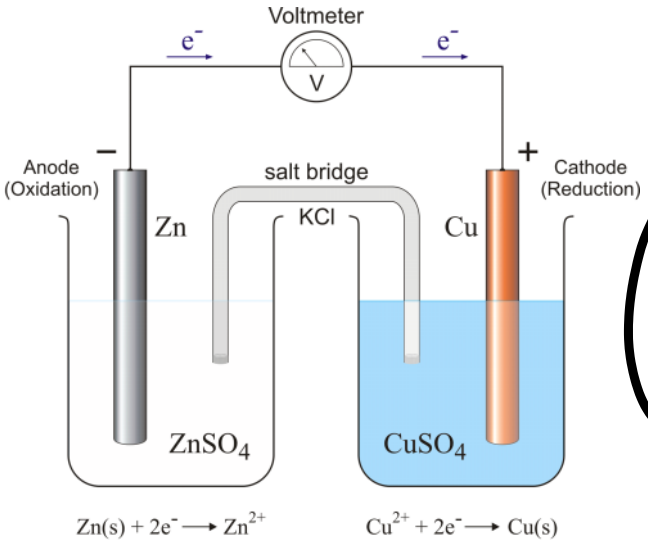
The goal of this project is to investigate more sustainable and safe alternatives to current batteries.
4 major batteries now:

- Alkaline: nontoxic, safe
    - Not rechargeable
    - Lower shelf life due to internal reactions
- Ni-MH: Safe and powerful, nontoxic
    - High rate of energy dissipation when not in use
- Lead-acid: lasts a very long time, works in any weather, for electronics that won't be touched often
    - Super toxic
- Li-ion: Powerful, rechargeable, powers all rechargeable electronics
    - Has a flammable organic solvent

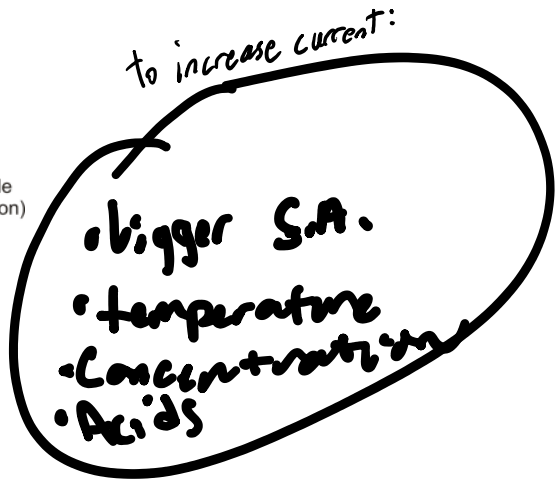
None are safe and long-lasting (as in low self-discharge, low internal reactions, rechargeable, and hopefully sustainable), so maybe we can find something that is better for that

First Cell

Tuesday, September 17, 2024

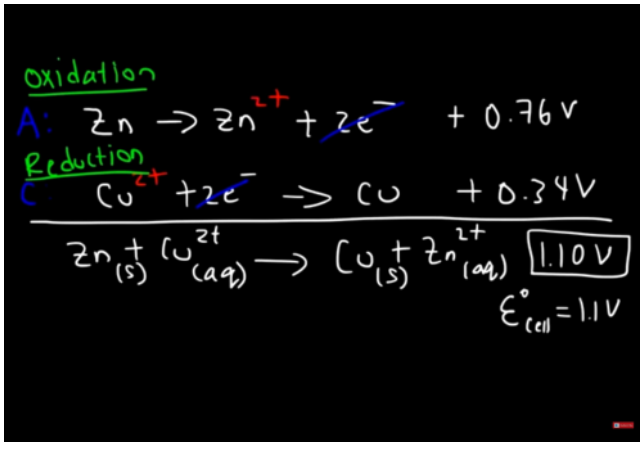


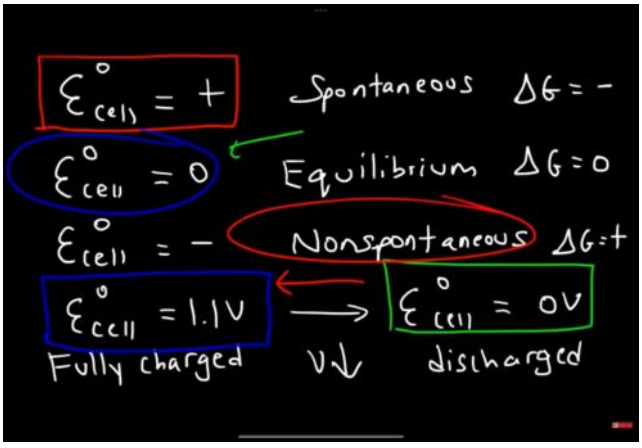
$$Zn(s) \mid ZnSO_4(aq) \,||\, CuSO_4(aq) \mid Cu(s)$$



Resources we are provided: Zinc and copper metals and sulfates

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
$Cu(OH)_2 + H_2SO_4 \rightarrow CdSO_4 + H_2O + CO_2$
$CdCO_3 + H_2SO_4 \rightarrow CdSO_4 + H_2O + CO_2$





Materials:

- Zinc (metal)
- Copper (metal)
- ~~Sulfuric Acid~~ We have the sulfates
- Beaker
- Salt Bridge (KCl)
- some sort of load for the system

~~$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$~~
~~$Cu + 2H_2SO_4 \rightarrow CuSO_4 + 2H_2O + SO_2$~~

Cell Formula: $Zn(s) \mid ZnSO_4(aq) \,||\, CuSO_4(aq) \mid Cu(s)$

Procedure:

1. Use sulfuric acid to create zinc and copper (II) sulfates in water
2. Pour each sulfate into a beaker
3. Insert respective metal electrode into aqueous solution, with wire attaching to each
4. Set up the load by connecting it to the electrodes with wires

5. Insert KCl salt bridge and watch the reaction take place

$Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ … (Open circuit voltage: 1.1018 V)

Cu: 21.70 g - Zn: 8.73 g

$$0.100\ L \times \frac{1\ mol}{L} \times \frac{287.59\ g\ ZnSO_4 \cdot 7H_2O}{mol} = 28.76\ g\ ZnSO_4 \cdot 7H_2O$$

Zinc Sulfate Heptahydrate

$$0.100\ L \times \frac{1\ mol}{L} \times \frac{159.61\ CuSO_4 \cdot 5H_2O}{mol} = 15.96\ g\ CuSO_4 \cdot 5H_2O$$

Copper (II) Sulfate Pentahydrate

Secondary batteries

Chemistry	Cell voltage	Specific energy (kJ/kg)	Energy density (kJ/liter)	Comments
Lithium ion	3.6	460		Very expensive. Very high energy density. Not usually available in "common" battery sizes. Lithium polymer battery is common in laptop computers, digital cameras, camcorders, and cellphones. Very low rate of self-discharge. Terminal voltage varies from 4.2 to 3.0 volts during discharge. Volatile: Chance of explosion if short-circuited, allowed to overheat, or not manufactured with rigorous quality standards.
AgZn	1.86 1.5	460		Smaller volume than equivalent Li-ion. Extremely expensive due to silver. Very high energy density. Very high drain capable. For many years considered obsolete due to high silver prices. Cell suffers from oxidation if unused. Reactions are not fully understood. Terminal voltage very stable but suddenly drops to 1.5 volts at 70–80% charge (believed to be due to presence of both argentous and argentic oxide in positive plate; one is consumed first). Has been used in lieu of primary battery (moon buggy). Is being developed once again as a replacement for Li-ion.
NiZn	1.6	360		Moderately inexpensive. High drain device suitable. Low self-discharge rate. Voltage closer to alkaline primary cells than other secondary cells. No toxic components. Newly introduced to the market (2009). Has not yet established a track record. Limited size availability.
$LiFePO_4$	3.3 3.0	360	790	Lithium–Iron–Phosphate chemistry.
NiMH	1.2	360		Inexpensive. Performs better than alkaline batteries in higher drain devices. Traditional chemistry has high energy density, but also a high rate of self-discharge. Newer chemistry has low self-discharge rate, but also a ~25% lower energy density. Used in some cars.
Lead–acid	2.1	140		Moderately expensive. Moderate energy density. Moderate rate of self-discharge. Higher discharge rates result in considerable loss of capacity. Environmental hazard due to Lead. Common use: automobile batteries
NiCd	1.2	140		Inexpensive. High-/low-drain, moderate energy density. Can withstand very high discharge rates with virtually no loss of capacity. Moderate rate of self-discharge. Environmental hazard due to Cadmium, use now virtually prohibited in Europe.

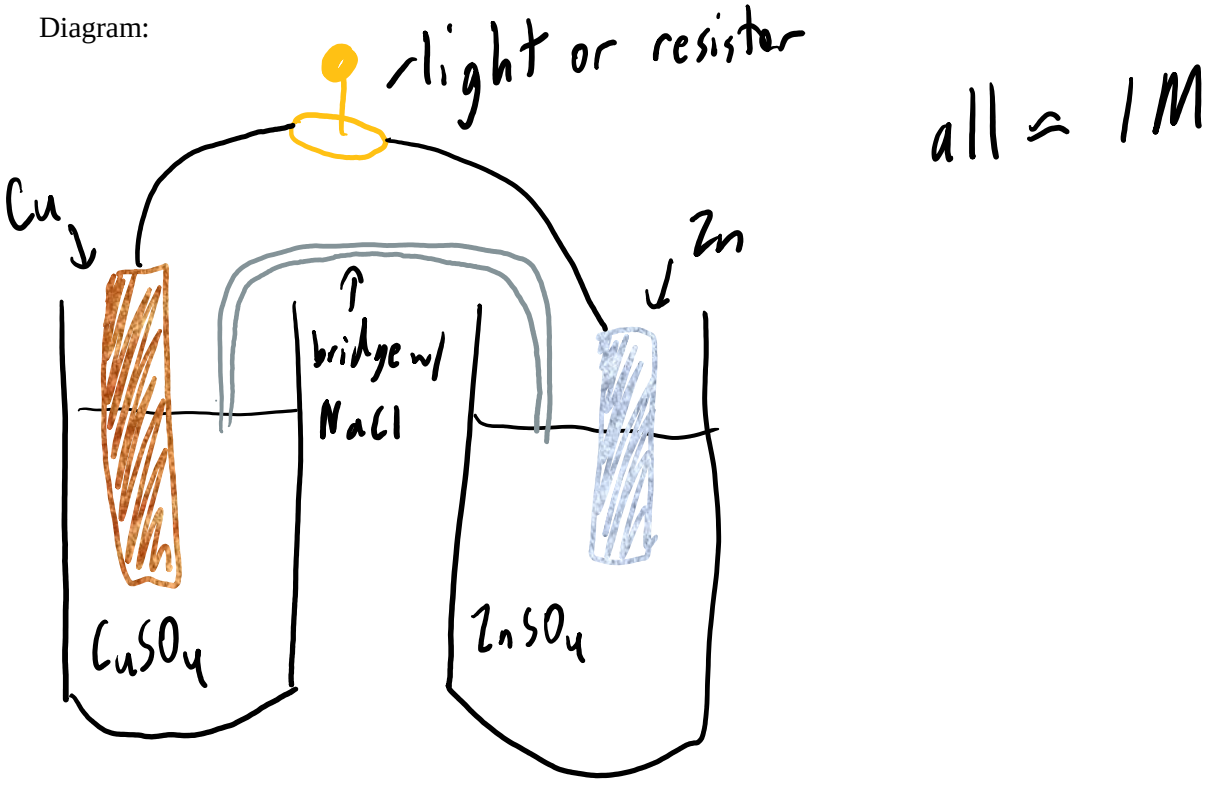
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Basic Cell: Further Testing

Monday, September 23, 2024

Created a voltaic cell using zinc, zinc sulfate, copper, and copper sulfate
Cu: 21.70 g - Zn: 8.73 g

Diagram:



Tested voltage
1.02 V with no resistance
0.52 V with 1000 ohm resistor
= so about 1000 ohms internal resistance: very high

Tested current
About .8 mA with no resistance
A little under the 1.1 mA predicted
Current varied with level of NaCl in bridge
Heat increased the current a bit

Lightbulb could not turn on, too much resistance so very low voltage

Things learned:

Concentration, heat, and SA are the biggest factors for resistance (and so current)

- [NaCl] allows for transfer of ions across bridge, which was the limiting factor
- Heat = more redox reactions = more e^- released
- More SA means more reactions

Need to find ways to lower resistance and increase current

- 1 V is good, but $V = IR$, so the high resistance is ruining current
- To change V, we will use other materials, but not yet

Most batteries have only one electrolyte, not two like ours

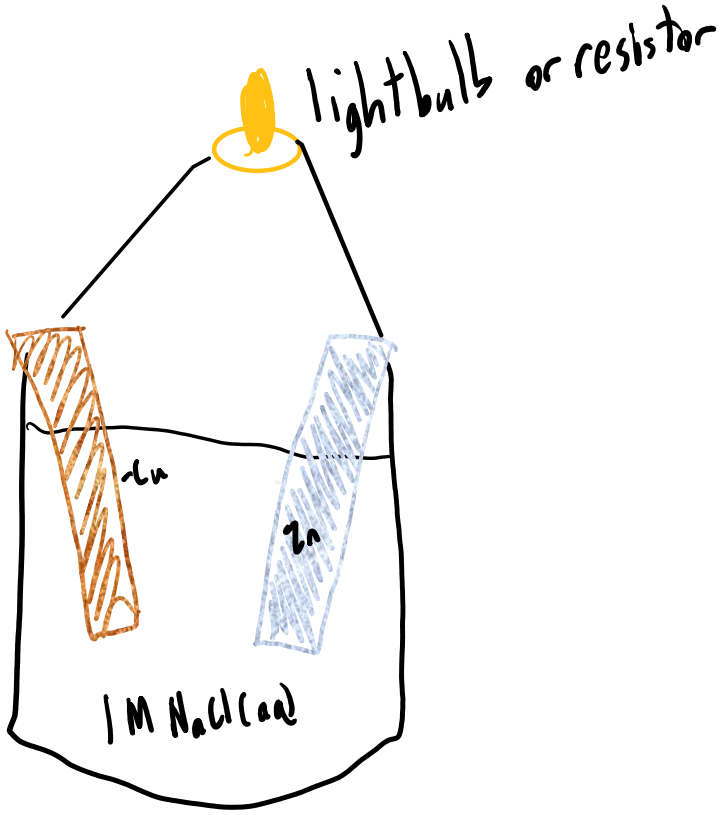
- This means no salt bridge, but the electrodes must not react with electrolyte without releasing charge
- This will also lower resistance, especially if solvent is conductive

Goal:

- find ways to lower R without super high concentrations
- Make a battery using only one electrolyte
- Try to find a reversible reaction if possible (it's ok if we can't), so we can make a rechargeable battery

Voltaic Cell in 1 Beaker

Wednesday, September 25, 2024



Measured voltage with 1000 ohm resistor: 0.8-0.9 V
Measured voltage with a 100 ohms resistor: 0.4 V

Used resistance box for variable resistance
Used 1 MΩ resistance to test cell potential, got 0.9 V

Since it was about half with 100 ohms, the internal resistance is 100 ohms
Current: $V = IR, I = \frac{V}{R}, I = \frac{0.9\ \text{V}}{100\ \Omega} = 9\ \text{mA}$

Redid experiment with 2 M NaCl concentration
New resistance was 50 ohms
$\frac{1}{2}R = 2I, I = 18\ mA$

Last attempt at 4 M NaCl, V decreased
Possible reasons:

- Max reaction rate of the zinc and copper, and concentrations were going down
- Excess sodium ions getting in the way?

Reactions:
Anode (Zn): $\text{Zn(s)} \rightarrow \text{Zn}^{2+}\text{(aq)} + 2\text{e}^{-}$
Cathode (Cu):

- There isn't any copper to reduce
- Sodium is essentially impossible to reduce

- So it must be water

$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$

But this reaction is highly unfavorable (-0.8 V)

- Here, the hydrogen is being reduced, which is unfavorable
- Reduction of sodium and chlorine is obviously highly unfavorable

There should be a slow production of hydrogen ions

Solution should be turning basic over time

- next thing to test

0.9 V = ϵ

100 Ohms, resistance (because voltage halves at 100 ohms from box)

with 1 M NaCl (*aq*) as electrolyte

50 ohms in 2 M NaCl

Lowered V at 4 M

Possible reasons:

hitting max reaction rate of the zinc and copper, this is only limited by the surface area of the electrodes

Maybe excess sodium ions are getting in the way?

Zinc is the stronger reducing agent, copper reduces and zinc oxidizes

Anode (Zn metal): $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ 0.78 V

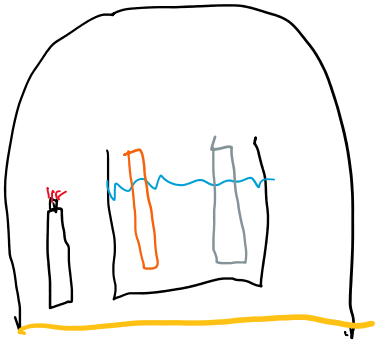
Cathode:

- Can't be zinc since that is the anode
- There aren't any copper ions to reduce
- Sodium reduction is extremely unfavorable and is almost guaranteed not to happen
- Leaves only water

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

$2H_2O + 2e^- \xrightarrow{Cu} H_2 + 2OH^-$

Possible reductions



$2Cl^- \rightarrow Cl_2 + 2e^-$ ≈ -1.78 V

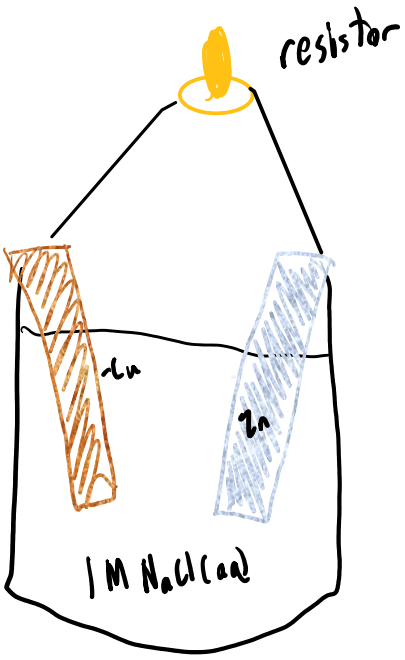
$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ ≈ 0.40 V

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ ≈ 0.40 V

~~$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$~~ ≈ −0.83 V

Identifying the Reaction

Monday, September 30, 2024



On Friday, we recreated our cell from Wednesday
Today, we noticed the white precipitate again
pH: 7, completely neutral (should have been basic by our predictions)

- Means there was no dissolved hydroxide
- However! The white precipitate could be from reaction!

Most likely reaction:
~~$Zn(s) + O_2 + H_2O \rightarrow Zn(OH)_2(s) + 2OH^-$~~

- ~~This would imply that the reaction is turning basic over time~~
  ~~But it didn't!~~

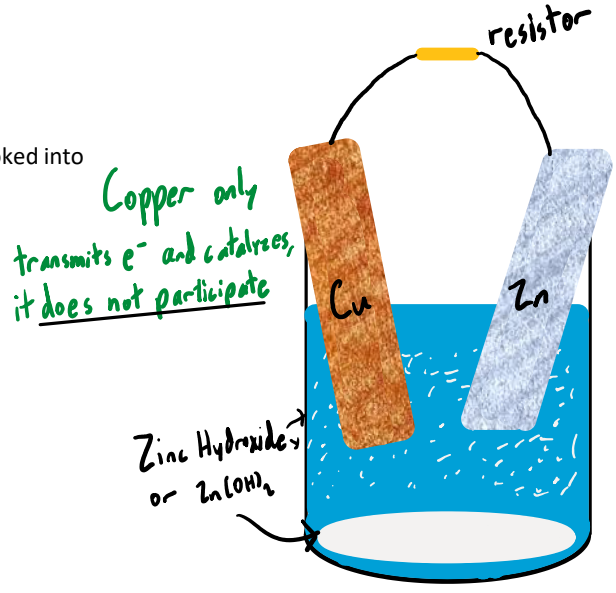
$Zn(s) \rightarrow Zn^{2+} + 2e^-$
$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
Overall reaction: $2\,\mathrm{Zn}(s) + \mathrm{O_2}(g) + 2\,\mathrm{H_2O}(l) \rightarrow 2\,\mathrm{Zn(OH)_2}(s)$

- Thus, no hydroxide ions are being formed
- Zinc Hydroxide will precipitate out though, confirming observations
  - Need to try to limit side reactions of this nature
  - Questions
    - Does zinc hydroxide still form when battery isn't running?
    - Is zinc hydroxide formation reversible (aka electrolytic cell)?
- Besides zinc hydroxide, the reaction is reversible
  $Zn(s) + O_2 + H_2O \rightarrow Zn^{2+} + 4OH^-$ (produces around 1.1 V theoretical)
  $Zn^{2+} + 4OH^- \rightarrow Zn(s) + O_2 + H_2O$ (about 1.1 V needed, but O_2 has high overvoltage)
  - Need to investigate overvoltage as well
- More research is needed

left over the weekend

NEXT STEPS:

- Test other electrodes (aluminum, magnesium, etc.)
  - Unused electrodes like iodine and graphite should also be looked into
- Look into other electrolytes
  - Conductive liquids?
    - Mercury: will create zinc amalgam, so it doesn't work
    - Gallium: needs much higher temperatures
    - Something organic? (I don't know much about this)
  - I don't think other dissolved solutes would work better
    - Sodium chloride is already highly soluble
    - Maybe replace copper with an
- New reduction/cathode reaction
  - Find something better to reduce so hydroxide doesn't form
    - Side reactions will kill battery life
- Begin writing procedure
  - Already have conducted an experiment
  - Talk about what electrodes we might use
  - Explain the difference between conductive liquids and dissolved solutes



Reaction Research

Tuesday, October 1, 2024

Zinc–air battery

Article Talk

From Wikipedia, the free encyclopedia

A **zinc–air battery** is a metal–air electrochemical cell powered by the oxidation of zinc with oxygen from the air. During discharge, a mass of zinc particles forms a porous anode, which is saturated with an electrolyte. Oxygen from the air reacts at the cathode and forms hydroxyl ions which migrate into the zinc paste and form zincate ($Zn(OH)_4^{2-}$), releasing electrons to travel to the cathode. The zincate decays into zinc oxide and water returns to the electrolyte. The water and hydroxyl from the anode are recycled at the cathode, so the water is not consumed. The reactions produce a theoretical voltage of 1.65 Volts, but is reduced to 1.35–1.4 V in available cells.

These batteries have high energy densities and are relatively inexpensive to produce. Zinc–air batteries have some properties of fuel cells as well as batteries: the zinc is the fuel, the reaction rate can be controlled by varying the air flow, and oxidized zinc/electrolyte paste can be replaced with fresh paste.

Screen clipping taken: 10/1/2024 11:29 AM

This is what we had made
So instead of making zinc hydroxide, it makes zincate to zinc oxide

- Zinc oxide is insoluble, which makes sense

Reaction formulas [edit]

The chemical equations for the zinc–air cell are:[2]

Anode:

$$Zn + 4\,OH^- \longrightarrow Zn(OH)_4^{2-} + 2\,e^- \quad (E_0 = -1.25\ V)$$

Fluid:

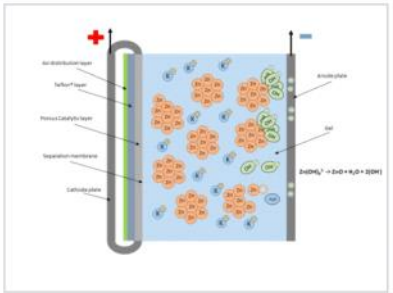
$$Zn(OH)_4^{2-} \longrightarrow ZnO + H_2O + 2\,OH^-$$

Cathode:

$$\tfrac{1}{2}O_2 + H_2O + 2\,e^- \longrightarrow 2\,OH^- \quad (E_0 = 0.34\ V\ pH = 11)$$

Overall

$$2\,Zn + O_2 \longrightarrow 2\,ZnO \quad (E_0 = 1.59\ V)$$



Animation of the operation of a zinc–air cell

Zinc–air batteries cannot be used in a sealed battery holder since some air must come in; the oxygen in 1 liter of air is required for every ampere-hour of capacity used.

Screen clipping taken: 10/1/2024 1:01 PM

Obviously we don't want our battery to need to be exposed to air, so we should look into something else

- Zinc-air batteries have also been made

Needs to be:

- Safe
- Rechargeable
- Hopefully rather cheap

Possible battery types:

- ~~Nickel-Zinc~~
  - ~~Well established, already made, not as much room to experiment~~
- ~~Magnesium-ion~~
  - ~~May not require intercalation~~
    - ~~Needs a suitable electrolyte to not require intercalation, and nobody has found one in 30 years~~
  - ~~+2 oxidation state creates more electrostatic forces, which is a problem for intercalation~~
- ~~Solid-State~~
  - ~~Way too expensive (tens of thousands of dollars), we are definitely NOT doing this~~
- Zinc-ion
  - Requires intercalation
  - Bigger radius than lithium and +2 oxidation state makes intercalation harder
- ~~Sodium-ion~~
  - ~~Requires intercalation~~
  - ~~HIGHLY reactive, because of sodium~~
  - ~~Too dangerous~~
- ~~Calcium-ion~~
  - ~~Requires intercalation~~
  - ~~Highly experimental, very little research has been done~~
  - ~~We don't have the money to do the research~~
- ~~Iron-air~~
  - ~~Well established, already made, not as much room to experiment~~
- ~~Polymer-based~~~~?~~
  - ~~Seems cool :)~~
  - ~~Very expensive~~

To recap:

- Magnesium-ion
- Zinc-ion
- & Polymer-based

are our best options.

Exploring Battery Types

Wednesday, October 2, 2024

This is a rundown and explanation of the types of batteries that matter for this project, with my (Nirvan) thoughts included.

1. **Zinc-air batteries:**
    a. Advantages:
        i. Higher energy density because air is one of the reactants, and doesn't take up space
        ii. Long shelf life when sealed
        iii. Stable voltage up until complete battery discharge
    b. Disadvantages:
        i. Needs air, so it cannot be sealed while in usage
        ii. Battery is dependent on ambient conditions
        iii. Horrendous cycle efficiency as low as 50%
        iv. Zinc corrosion can cause explosive hydrogen gas buildup
        v. Zinc forms dendrites
        vi. Once its started, it (basically) can't be stopped
    c. Application
        i. Vehicle propulsion
        ii. Grid storage
    d. Decision: **NO**, because it needs air and is bad for rechargeable
2. **Zinc-ion**
    a. Pros:
        i. Safe and non-toxic
        ii. Low cost and abundant materials
        iii. Safer to handle
        iv. Potential for high energy density
    b. Cons:
        i. Dendrite formation
        ii. Side reactions limit cycles
        iii. Hard to find good cathode
    c. Applications:
        i. Grid energy storage
        ii. Emergency power supplies
        iii. Electric vehicles
        iv. Portable electronics
3. **Magnesium-ion**
    a. Pros:
        i. Safe and non-toxic
        ii. Low cost and abundant materials
        iii. Potential for even higher energy density than zinc-ion
        iv. Less likely for dendrite formation
    b. Cons:
        i. Hard to find an electrolyte
        ii. Hard to find compatible cathode materials
        iii. Slower ion-diffusion (= lower current) vs. lithium-ion
    c. Applications:
        i. Electric vehicles
        ii. Grid energy storage

iii. Portable electronics

4. **Polymer-based**
   a. Pros:
      i. Can be safe
      ii. Can be biodegradable
      iii. Many flexible designs possible for many applications
   b. Cons:
      i. Lower energy density
      ii. May require complex polymers
   c. Applications:
      i. Flexible electronics
      ii. Wearable technology

CURRENT PLAN: POLYMER BATTERIES & ZINC-ION BATTERIES
Because…

Polymer Battery:

Biodegradable (= sustainable & safe)
- This replaces long-lasting, because if it stops working, it won't harm anything, as it just breaks down into organic substances

Rechargeable (= sustainable)
- In fact, the recharge rate can reach ridiculously high levels, due to the absence of intercalating metals (there is a study of an organic radical battery completely recharging in 6 seconds)

Flexible (= our application)
Safe and non-toxic with right material choices (= safe)
Lots of area for exploration (= our research)
Only major problem is lower energy density, which isn't our focus
Only worries are cost and time needed to design the battery.

Zinc-ion Battery:

~~The other contender was Mg-ion battery. But, Mg-ion is much more complex.~~
~~Mg-ion:~~
- ~~Needs an electrolyte to facilitate magnesium transport, but conventional electrolytes degrade on contact with magnesium metal~~
- ~~Does not need to be intercalated~~

Zn-ion:
- Any electrolyte (including aqueous) works, allowing for much safer batteries
- However, it does need to be intercalated, but apparently this is not expensive.
- Higher theoretical energy density than even lithium
- Higher recharge rate (nothing compared to polymer battery)
- Extremely safe, and should last for very high number of cycles

NEXT STEPS:
- Contact Mr. Green about conductive polymers, creating and designing polymers, and manipulating polymers as well.
- Study some organic chemistry and learn about the composition of polymers
- Start making a zinc-ion battery (much simpler)
  - Graphite, zinc, and some casing is pretty much it
  - Homemade 1 Amp Hour Zinc Ion Battery - Full Instructions
  - Zinc Ion High Energy Home Made Battery - The Smallest 1Ah Cell I've Made..

↑ This guy straight up just makes DIY Zinc-ion Batteries in his spare time.

Homemade Zinc-ion Battery

Friday, October 18, 2024

- Homemade 1 Amp Hour Zinc Ion Battery - Full Instructions
- Zinc Ion High Energy Home Made Battery - The Smallest 1Ah Cell I've Made..

Notes on how to design:

Cathode:

- Glue paper into cylindrical oval housing
  - To hold the carbon
- Insert a strip of graphite rod into the paper
  - Current collector
- Measure out:
  - 11g of carbon powder
  - 2g of graphite powder
    - For conductivity
- Stir together
- Add a small amount of zinc sulfate sodium sulfate
  - Make by combining 1 M zinc and sodium sulfates
  - Electrolyte
- Mixture should be damp but not wet
- Once mixed, shovel the stuff into the paper
- Use pencil to pack it down firmly
- Seal the top with hot glue
- Pack together with yarn

Anode:

- All that is needed is zinc
- Space the zinc using something like plastic mesh
  - Stops zinc dendrites from shorting
  - I guess not technically necessary

- Top up with electrolyte
- Charge the battery with 2.2 V

The Beaker Battery

Tuesday, October 22, 2024

NOT USED FOR DATA COLLECTION

We used a 400 mL beaker to build the battery in.
First, the graphite rod was put in the beaker
Next, we made the electrolyte mixture

- Total:
    - Roughly 33g carbon powder
    - About 6g graphite powder
    - Around 30 mL of 1M aqueous Zinc Sulfate + Sodium Sulfate
    - A graphite rod
    - A zinc strip

The powder and electrolyte made a "goop", which was then shoveled into the beaker.
The 11g carbon/2g graphite powder mixture wasn't enough, so we tripled it

Produced 1.2V
However, that dropped to about .1V with a .6 ohm lightbulb
Internal Resistance: $7\ \Omega$

- That a 93% decrease from the last cell
- Why? We don't know yet
    - 1 M Zinc Sulfate + Sodium Sulfate is isoelectronic to 2 M zinc sulfate, but the old cell had 3 M NaCl
    - Ions move at around the same speed, so a single ion holding more charge is more efficient
        - Thus, SO_4^{2-} is more efficient than Cl^-, and Zn^{2+} is more efficient than Na^+
- However, most batteries have resistances in the hundredths of Ohms

Current Produced: $V = IR, I = \frac{V}{R}, I = \frac{0.1V}{7.6\Omega} \approx 13.16\ \text{mA}$, which is about what we measured (11 mA measured)
So, we need to get that resistance down a lot.

We were able to charge the battery: it drew .62 A at around 2 V, which is a good charging speed
Max voltage was 1.8 V after charging for 10 mins, the actual maximum voltage (research) is around 2.2 V

This was very good, even though we forgot to add as much electrolyte as we were supposed to.
NEXT STEP: Add more electrolyte, try changing electrolyte and carbon mixtures, etc.

The Big Battery

Wednesday, October 30, 2024

NOT USED FOR DATA COLLECTION

We are now using our own container for the battery, 3D printed from ABS plastic

DXF diagram:

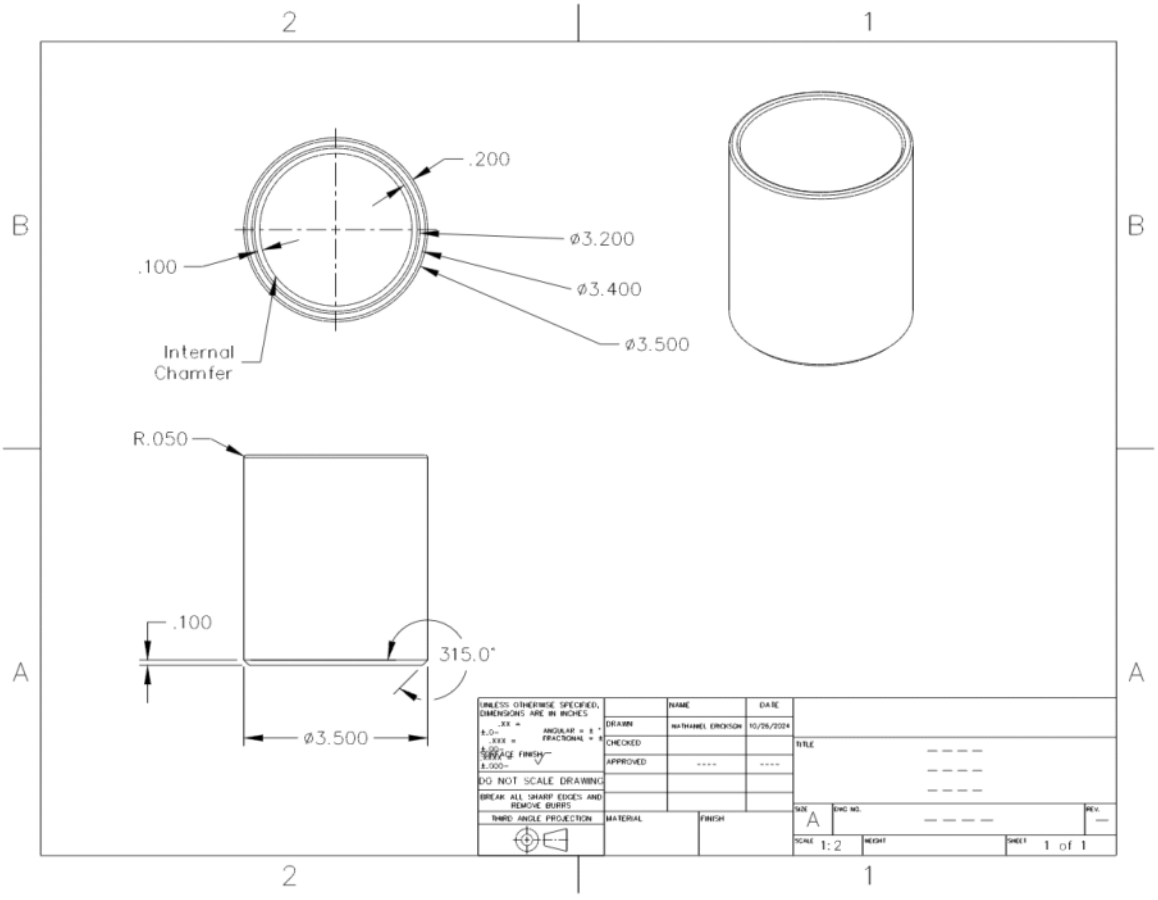


Figure 1: A DXF diagram of the design for the internal casing. All lengths are given in inches and all angles are given in degrees.

It is a 4-inch tall, 1.75 inch radius cylinder. We went through the same steps, but we hit the max solubility of zinc and sodium sulfates.

Materials used:

- 33g carbon powder (affects capacity)
- 6g graphite powder (affects internal resistance and capacity)
- 200 mL of 1.5M Zinc Sulfate and Sodium Sulfate (affects internal resistance)
- Graphite rod (cathode)
- Zinc strip (anode)

Once again, we made a "goop" that we put in the container, but now we filled the container with electrolyte so the conductivity would be much higher.

As of now, the main issue is too high internal resistance. The ways to decrease this are:

- Concentration of electrolyte
- Surface area of electrodes
- Temperature

We will test the battery in a circuit tomorrow.

This battery got even worse

Round 3

Monday, November 11, 2024

NOT USED FOR DATA COLLECTION

We still have many mistakes…

The graphite is surrounded by the powder mixture <u>within its own casing</u>

- This casing is tightly sealed

The zinc is outside of this casing, but electrolyte floods the two electrodes
Plastic mesh between the graphite and zinc

At 2:40 pm, round 3 battery is done.
Cathode: 20 g carbon powder, 8 g graphite powder, sealed inside cathode casing with graphite rod sticking out
Anode: the zinc strip
Both are placed in a 100 mL beaker (to minimize volume and maximize surface area)
The beaker is then topped up with as much electrolyte as possible (50 mL)

Data collection time

Friday, December 20, 2024 12:04 PM

NOT USED FOR DATA COLLECTION

Added 39.23 g $ZnSO_4$ to 80mL water for about 3M solution of around 100 Ml

Added 15.56 g activated charcoal (carbon)
Added 6.33g graphite
Mixed together in bowl, combined with 25 mL of electrolyte

Smushed inside graphite foil
Wrapped in filter paper and placed in casing with zinc metal and electrolyte
Did not work, probably because of the graphite foil

Finally, it works

Saturday, December 21, 2024 11:53 AM

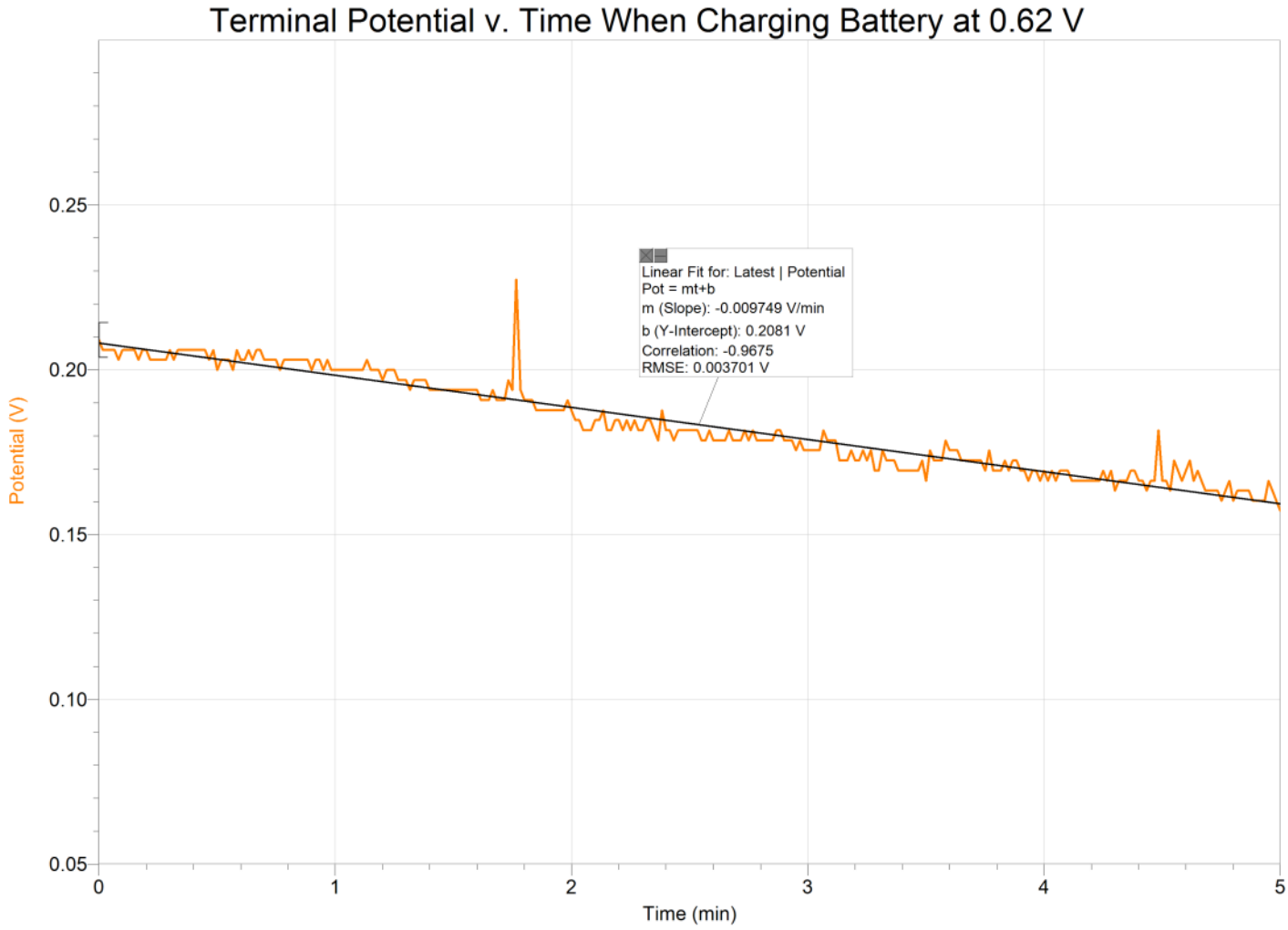
Battery works today
Replaced graphite foil with graphite rod
Gave 1.2 volts
We are charging it, then we will do a discharge test

Using power supply as emf, terminals as V, and power supply's current as I,
$V = \varepsilon - \mathrm{IR}$
Gives the internal resistance as:
At 1 V, .27 A charge rate: $1.574\ \Omega$
At 2 V, .66 A charge rate: $1.280\ \Omega$
At 2.49 V, 1.04 A charge rate: $1.012\ \Omega$

Why is resistance going down with current?
After charging, there is no voltage across the battery
We may have discharged it by accident

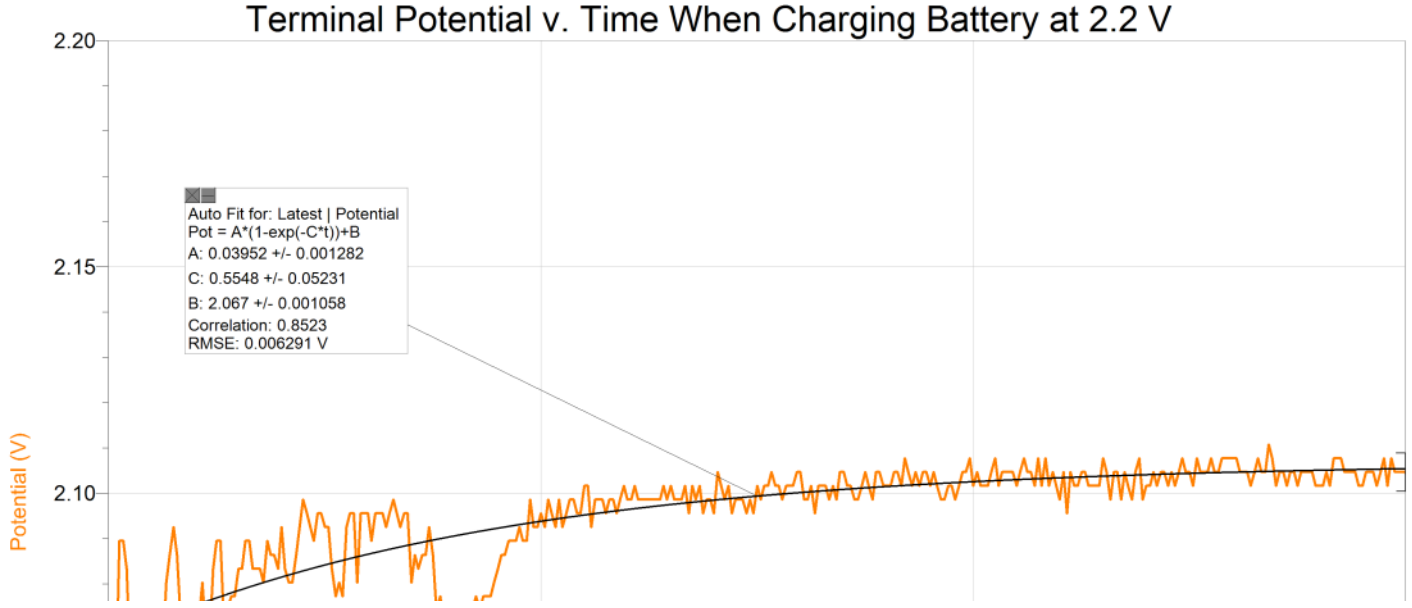
Flipping the current seemed to resolve the problem

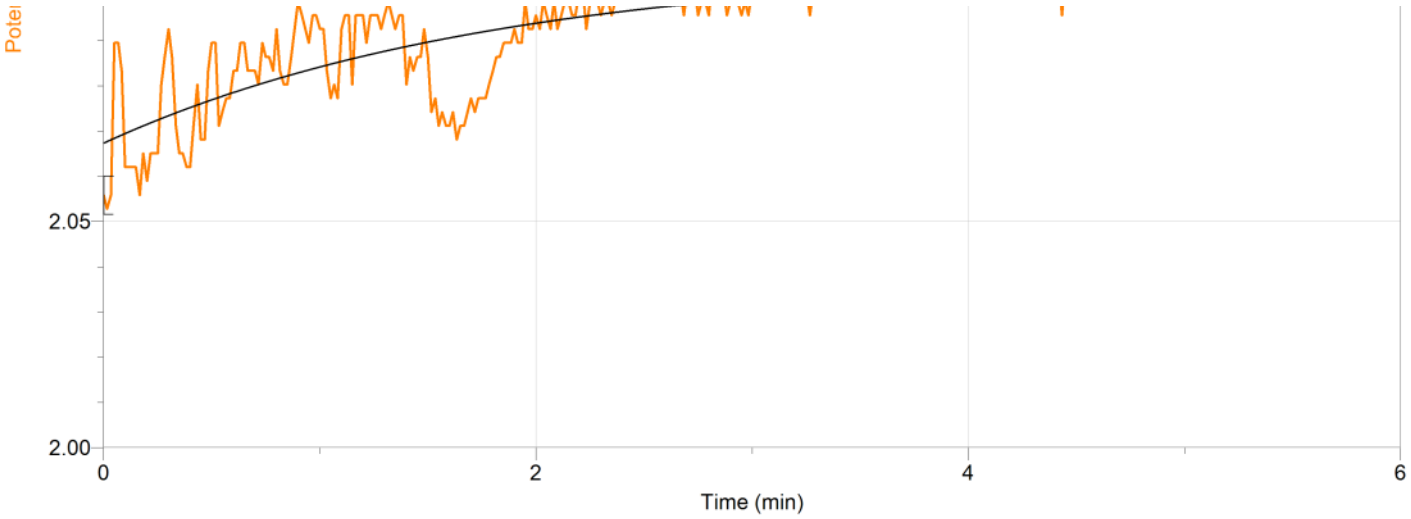
When the cathode and the anode touch, then the current that the power supply reads increases, on the other hand, when the two are separated, then the current read is 0



This time, we folded the zinc and made sure that it does not touch the cathode's filter paper separator

- We guess that the weird trend was caused by the battery's electrodes touching
- Thus, the filter paper is cannot keep the electricity from going straight from electrode to electrode
- We were able to charge the battery and create a graph of the potential, which seemed like an inverse exponential





We cannot find the internal resistance because we do not have a resistor.

Round 2

Saturday, December 28, 2024 5:26 PM

FIRST TEST WITH DATA COLLECTION

15.56 g charcoal
6.76 g graphite
25 mL electrolyte

1 OHM INTERNAL RESISTANCE!!!

1.060 v across battery to begin (starting emf)
11.25 Ω internal resistance measured

In second discharge test:
Starting emf: 1.28 V, Internal resistance at beginning was 4.372 Ω
ending values: 0.361 V, 10.755 Ω

Time to recharge:

Stuff for fair

Tuesday, December 31, 2024 12:47 PM

- Test ni-mh and li-ion batteries
- Test capacitors
- Try zinc chloride and other, lighter things

Making the second battery

Wednesday, January 8, 2025 2:38 PM

3.74 g graphite
9.22 g activated carbon
22.3 g zinc sulfate in 50 mL water for 3M concentration
This time, the battery will be charged and discharged for longer. The graphite and activated carbon have been both halved, so the discharge curves are expected to hold half the charge

Charging plan

Thursday, January 9, 2025 5:23 PM

Emf can't be measured while battery is charging
But, if the battery is momentarily disconnected, the emf can be measured
So, a circuit that automatically disconnects, measures the emf, then reconnects could work to measure emf effectively

Materials:

- The battery
- The power supply
- Voltage probe
- A relay (aka electromagnetic switch)
- A microcontroller (in this case, an arduino)

1. Hook the battery up to charge, with voltage probe across the battery terminals and a relay in the circuit
2. Use the arduino to activate the relay after 4 minutes, leave the circuit open for a minute, then switch the relay back again, continuously.
3. Have logger pro pick up the data every 5 minutes, set so it collects the emf while the circuit is open.

This circuit will allow for a better measurement of the emf

How to calculate internal resistance while running?

- Could use a second relay and momentarily measure the current and voltage across a known resistance to find the internal resistance (very complicated)
- By Kirchhoff's Second Law, $V_{supply} - \varepsilon - Ir = 0$, or $r = \frac{V_{supply} - \varepsilon}{I}$, all of which can be calculated during the experiment

How to calculate the current?

Discharge Circuit

Tuesday, January 14, 2025 10:02 PM

Materials for discharge circuit:

Volt probe
Current probe
Resistance box
Battery

Procedure

1. Wire the voltage probe across the the battery
2. Wire the resistance box across the battery, with the current probe in series (disconnect circuit though to prevent discharge)
3. Connect current probe and voltage probe to labquest
4. Change labquest to sample 6/min for 1080 min
5. Reconnect circuit, then leave labquest running

Zinc Chloride tests

Thursday, January 16, 2025 11:57 PM

We are replacing zinc sulfate with zinc chloride

1. Zinc chloride used in cathode and electrolyte with a graphite rod
    a. Can also be replaced with zinc sulfate electrolyte
2. New graphite and carbon amounts
    a. 9.88g carbon
    b. 11.19g graphite
    c. Zinc chloride electrolyte
3. Graphite foil battery, using old carbon/graphite concentrations and zinc chloride

Anode will forever be constant: zinc metal

Iterations List

Friday, January 17, 2025 10:27 AM

Iteration 1: graphite rod, zinc sulfate goop, zinc sulfate electrolyte
Iteration 2: half the goop (carbon powders) of iteration 1
Iteration 3: Graphite rod, zinc chloride goop, zinc chloride electrolyte
Iteration 4: 1:1 ratio of powders from Iteration 3

Next:
Iteration 5: cathode 3 in zinc sulfate

Iteration 6: graphite foil, manganese oxide

16 g carbon
7.77 g graphite
6.47 g manganese dioxide

Used distilled water instead

86 g zncl2 to make 6 molar

Sadly, charging produced chlorine gas, and experimentation had to be shortened down

- Time to begin paper